

Francisco Rodríguez

Data & AI Engineer | Data Science | Machine Learning | Generative AI and Agentic Systems

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Professional Summary

Data and AI engineer with a scientific background and experience building data products from ingestion and analysis to production deployment. Combines data engineering, machine learning and product development to deliver reliable, governed solutions for technical teams and business users.

Professional Experience

Gran Telescopio de Canarias (GRANTECAN), Data & AI Engineer

07/2025 - 07/2026

- Developed an agentic RAG assistant using local LLMs to review and draft technical and administrative tender specifications, **reducing manual effort by about 50%** and grounding outputs in internal documentation.
- Developed a conversational data assistant as part of the company's first dedicated AI group using speech recognition, LangGraph and RAG, enabling employees without a technical background to query Delta Lake data in natural language and receive visual responses.
- Built PySpark ETL pipelines for the corporate on-premises data lake, filling data gaps spanning several months and delivering clean, normalized and aggregated datasets that **reduced the time needed to derive insights by about 40%**.
- Developed and maintained a React and Node.js application for managing telescope operating time and value statistics, replacing Excel and synchronous onsite coordination with a shared asynchronous workflow. The application **reduced a monthly process requiring 4 to 5 hours per contributor to about five minutes of daily upkeep**.
- Deployed local LLM inference infrastructure with vLLM, SLURM and RHEL, **serving concurrent users at about 200 tokens per second** and supporting internal projects that required full data sovereignty.
- Led the development of COMET, a human-in-the-loop operational platform for GTC, combining digital SOPs and runbooks with versioning, auditability, guardrails and AI-assisted workflows for controlled operational decision support.
- Co-authored the SPIE publication "[Cognitive artificial intelligence ecosystem with intelligent agents for GTC operations](#)", describing the integration of generative AI agents, operational intelligence, human validation and guardrailed orchestration across telescope workflows.

Cabildo de La Palma, Data Scientist

01/2024 - 01/2025

- Led EDA with Python and R across 5+ air-quality datasets in Puerto Naos, uncovering a **Pearson correlation of ~0.6 between coastal CO2 levels and tidal cycles**; the finding drove the Emergency Department's post-eruption habitability protocol review.
- Designed interactive dashboards in Power BI, Tableau and ArcGIS Online (including a **Puerto Naos digital twin** used daily by Emergency for volcanic CO2 tracking) and automated weekly Python workflows, **saving ~12 h per month** for non-technical stakeholders.
- Standardized and integrated a **3M+ record legacy dataset into PostgreSQL**, extending the Cabildo's accessible meteorological archive from 2022 back to 2016 and ensuring continuity for the public portal.

Additional experience: Private tutor in Mathematics, Physics and Chemistry.

Education

MSc in Data Science and Big Data, Open University of Catalonia (UOC)

2025

- Master's Thesis: sales forecasting and audience segmentation system for Filarmonía de Madrid with real ticketing data (Random Forest, CNN, RNN, ARIMA, K-Means, DBSCAN, RFM).

BSc in Physics, University of La Laguna (ULL)

2021

- Bachelor's Thesis: photometric calibration of the EMIR instrument (GTC, IAC) using Python and scientific documentation in LaTeX.

Skills

Programming: Python, SQL, JavaScript, TypeScript, R

Generative AI and ML: LLMs, RAG, agentic systems, LangGraph, LangChain, vLLM, embeddings, vector databases, NLP, PyTorch, TensorFlow, Scikit-learn

Data Engineering: PySpark, on-premises data lakes, medallion architecture, ETL, Apache Kafka, Trino, PostgreSQL, MySQL

Infrastructure and Deployment: Linux, RHEL, SLURM, Docker, Git, Nginx, Apache HTTP Server

Web and Visualization: React, Node.js, Power BI, Tableau, Grafana, ArcGIS Online, Matplotlib, Plotly

AI Development Tools: Claude Code, Codex, Cursor, OpenCode, agent configuration, tool integration, workflow automation

Other: Jinja, LaTeX

Languages: Spanish (native), English (fluent)